Problem Solving with Ratios

Exam code: 0607

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Working with Ratios

How do I share an amount in a given ratio?

- **Add** together **all parts in the ratio** to find the **total number of parts** in the whole
  - $200 is to be shared between two people, A and B, in the ratio 5:3
  - There are 8 "parts" in total, as A receives 5 parts and B receives 3 parts
- **Divide** the **amount being shared** by the **total number of parts**
  - $200 must be split into 8 parts, so this means that 1 part must be worth $25
    - 200 ÷ 8 = 25
- **Multiply** the **amount each part is worth** by the **number of parts** for each quantity in the ratio
  - Person A receives 5 parts, each worth $25
    - 5 × 25 = $125 for person A
  - Person B receives 3 parts, each worth $25
    - 3 × 25 = $75 for person B
- Check the **values** in the new ratio **add up to** the total **amount being shared**
  - $125 + $75 = $200

Examiner Tips and Tricks

- **Adding labels** to your ratios will help make your working clearer and help you remember which number represents which quantity, e.g. $\begin{matrix} A & : & B \\ 3 & : & 4 \end{matrix}$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

in a school.

Calculate the volume of red and white paint that Mark needs to purchase in order to have enough paint to decorate the room.

Answer:

The ratio of red to white is

$$3:2$$

Adding these together gives the total number of parts

$$3+2=5$$

The total amount of paint is 60 litres

$$\therefore 5 \text{ parts} = 60 \text{ litres}$$

Divide both sides by 5 to find out the number of litres in one part

$$\begin{array}{ccc} 5 \text{ parts} & = & 60 \text{ litres} \\ \div 5 & & \div 5 \\ 1 \text{ part} & = & 12 \text{ litres} \end{array}$$

The ratio was 3:2, so multiply both number of parts by 12

$$\begin{array}{ccc} \text{R} & : & \text{W} \\ 3 & : & 2 \\ \times 12 \downarrow & & \downarrow \times 12 \\ 36 & : & 24 \end{array}$$

Answer in context, making sure you make it clear which value is associated with which colour paint

Mark will need to buy 36 litres of red paint and 24 litres of white paint

What type of ratio problems could I be asked to solve?

- **Simple ratio problems** are discussed earlier, including
  - Writing ratios
  - The link between ratios and fractions
  - Equivalent ratios
  - Simplifying ratios
  - Sharing an amount in a given ratio
- **Further problems** involving ratio include
  - Ratios where you are given the **difference between** the **two parts**
    - E.g. Kerry is given $30 more than Kacey who is given $50
  - Ratios where **one quantity** is **given** and you have to **find** the **other quantity**
    - E.g. Kerry and Kacey are sharing money in the ratio 8 : 5, Kacey gets $50

- E.g. Kerry and Kacey are sharing money in the ratio 8 : 5 whilst Kacey is also sharing money with Kylie in the ratio 1 : 2

How do I solve a ratio problem when given the difference between two parts?

- **Find** the **difference in the number of parts** between the two quantities in the ratio
- **Compare** the difference in the **number of parts** with the difference between the **actual numbers**
- **Simplify** to find out the **value of one part**
- **Multiply** the **value of one part** by the **number of parts** for each quantity in the ratio
- **Multiply** the **value of one part** by the **total number of parts** to find the **total amount**

Given one quantity of a ratio, how can I find the other quantity?

- **Compare** the **given quantity** with the relevant **number of parts** in the ratio
- **Simplify** to find the **value of one part**
- **Multiply** the **value of one part** by the **number of parts** in the remaining quantity in the ratio
- **Multiply** the **value of one part** by the **total number of parts** to find the **total amount**

How do I combine two ratios to make a three-part ratio?

- **Identify** the **link** between the two different ratios
- **Find equivalent ratios** for both original ratios, where the **value** of the **link** is the **same**
- **Join** the two, two-part ratios into a **three-part ratio**

Worked Example

(a) The ratio of cabbage leaves eaten by two rabbits, Alfred and Bob, is 7 : 3 respectively. It is known that Alfred eats 12 more cabbage leaves than Bob for a particular period of time.

Find the total number of cabbage leaves eaten by the rabbits and the number that each rabbit eats individually.

Answer:

The difference in the number of parts is

$$7 - 3 = 4 \text{ parts}$$

This means that

$$4 \text{ parts} = 12 \text{ cabbage leaves}$$

Dividing both by 4

$$7 + 3 = 10 \text{ parts}$$

Find the total number of cabbage leaves

$$10 \times 3 = 30$$

30 cabbage leaves in total

Find the number eaten by Alfred

$$7 \times 3 = 21$$

21 cabbage leaves

Find the number eaten by Bob

$$3 \times 3 = 9$$

9 cabbage leaves

(b) A particular shade of pink paint is made using 3 parts red paint, to two parts white paint.

Mark already has 36 litres of red paint, but no white paint.

Calculate the volume of white paint that Mark needs to purchase in order to use all of his red paint, and calculate the total amount of pink paint this will produce.

Answer:

The ratio of red to white is

$$3 : 2$$

Mark already has 36 litres of red, so

$$36 \text{ litres} = 3 \text{ parts}$$

Divide both sides by 3.

$$12 \text{ litres} = 1 \text{ part}$$

The ratio was 3 : 2
Find the volume of white paint, 2 parts

$$2 \times 12 = 24$$

24 litres of white paint

In total there are 5 parts, so the total volume of paint will be

$$5 \times 12 = 60$$

60 litres in total

(c) In Jamie's sock drawer the ratio of black socks to striped socks is 5 : 2 respectively. The ratio of striped socks to white socks in the drawer is 6 : 7 respectively.

Calculate the percentage of socks in the drawer that are black.

Answer:

Write down the ratios

$$B : S = 5 : 2$$
$$S : W = 6 : 7$$

$$B : S = 15 : 6$$
$$S : W = 6 : 7$$

Link them together

$$B : S : W = 15 : 6 : 7$$

Find the total number of parts

$$15 + 6 + 7 = 28$$

This means 15 out of 28 socks are black
Find 15 out of 28 as a decimal by completing the division

$$\frac{15}{28} = 0.535\ 714\ 285\ \ldots$$

Convert to a percentage
Multiply by 100 and round to 3 significant figures

53.6 % of the socks are black

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